

T, TD - Solving trig - Solutions

Problem Solutions

- Solve these equations and verify your answers.
- If your answer is wrong, try to find your mistake and fix your solution.
- Afterwards, study the full solution provided.

Assume that x and θ lie in $[0, 2\pi)$ for all these problems.

1. $\cos(2x) - 2\cos x + 1 = 0$

Answer: $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

Solution >

Use $\cos(2x) = 2\cos^2 x - 1$:

$$\cos(2x) - 2\cos x + 1 = 0 \quad \ggg \quad 2\cos^2 x - 2\cos x = 0$$

Substitute $X = \cos x$ and solve:

$$\ggg \quad 2X^2 - 2X = 0 \quad \ggg \quad 2X(X - 1) = 0 \quad \ggg \quad X = 0, 1$$

- $\cos x = 1 \quad \ggg \quad x = 0$
- $\cos x = 0 \quad \ggg \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$

2. $\sin(2x) + \sin x = 0$

Answer: $x = 0, \pi, \frac{2\pi}{3}, \frac{4\pi}{3}$

Solution >

Use $\sin(2x) = 2\sin x \cos x$:

$$2 \sin x \cos x + \sin x = \sin x (2 \cos x + 1) = 0$$

So $\sin x = 0$ or $2 \cos x + 1 = 0$.

- $\sin x = 0 \quad \gg \gg \quad x = 0, \pi$
- $\cos x = -\frac{1}{2} \quad \gg \gg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

3. $\cos(2x) + \cos x + 1 = 0$

Answer: $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}$

 **Solution** >

Use $\cos(2x) = 2 \cos^2 x - 1$. Substitute $X = \cos x$ and solve:

$$\begin{aligned} \cos(2x) + \cos x + 1 = 0 & \gg \gg 2 \cos^2 x - 1 + \cos x + 1 \\ & \gg \gg 2X^2 + X = 0 \\ & \gg \gg X(2X + 1) = 0 \end{aligned}$$

So $X = 0$ or $X = -\frac{1}{2}$.

- $X = 0 \quad \gg \gg \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$
- $X = -\frac{1}{2} \quad \gg \gg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

4. $\cos(2x) + \frac{1}{2} = 0$

Answer: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

 **Solution** >

Use $\cos(2x) = 2 \cos^2 x - 1$ and solve for $\cos x$:

$$\cos(2x) + \frac{1}{2} = 0 \quad \ggg \quad 2\cos^2 x - 1 + \frac{1}{2} = 0$$

$$\ggg \quad 2\cos^2 x - \frac{1}{2} = 0$$

$$\ggg \quad \cos^2 x = \frac{1}{4}$$

$$\ggg \quad \cos x = \pm \frac{1}{2}$$

- $\cos x = \frac{1}{2} \quad \ggg \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$
- $\cos x = -\frac{1}{2} \quad \ggg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

5.

$$\tan(2\theta) - 3\tan\theta = 0$$

Answer: $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

 **Solution** >

Use $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$ and substitute $X = \tan\theta$:

$$\tan(2\theta) - 3\tan\theta = 0 \quad \ggg \quad \frac{2X}{1-X^2} - 3X = 0$$

$$\ggg \quad X\left(\frac{2}{1-X^2} - 3\right) = 0$$

Suppose that $X = 0$. Then $\theta = 0, \pi$.

Suppose that $\frac{2}{1-X^2} - 3 = 0$. Then:

$$\ggg \quad 2 = 3(1-X^2) \quad \ggg \quad X^2 = \frac{1}{3} \quad \ggg \quad X = \pm \frac{1}{\sqrt{3}}$$

- $\tan\theta = \frac{1}{\sqrt{3}} \quad \ggg \quad \theta = \frac{\pi}{6}, \frac{7\pi}{6}$
- $\tan\theta = -\frac{1}{\sqrt{3}} \quad \ggg \quad \theta = \frac{5\pi}{6}, \frac{11\pi}{6}$

6. $\sin^2 \theta + \sin \theta - 2 = 0$

Answer: $\theta = \frac{\pi}{2}$

 **Solution** >

Substitute $X = \sin \theta$ and solve:

$$\gggg \quad X^2 + X - 2 = 0 \quad \gggg \quad (X + 2)(X - 1) = 0$$

- $\sin \theta = -2 \quad \gggg \quad \text{impossible}$
- $\sin \theta = 1 \quad \gggg \quad \theta = \frac{\pi}{2}$

7. $\sin(2\theta) = \cos \theta$

Answer: $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

 **Solution** >

Use $\sin(2\theta) = 2 \sin \theta \cos \theta$:

$$2 \sin \theta \cos \theta = \cos \theta \quad \gggg \quad \cos \theta (2 \sin \theta - 1) = 0$$

- $\cos \theta = 0 \quad \gggg \quad \theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- $\sin \theta = \frac{1}{2} \quad \gggg \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

8. $\cos(2\theta) - \sin \theta = 0$

Answer: $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

 **Solution** >

Use $\cos(2\theta) = 1 - 2 \sin^2 \theta$ and substitute $X = \sin \theta$:

$$\ggg 1 - 2\sin^2\theta - \sin\theta \ggg 2X^2 + X - 1 = 0$$

Solve this using the quadratic formula:

$$X = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4} = \frac{1}{2}, -1$$

- $\sin\theta = \frac{1}{2} \ggg \theta = \frac{\pi}{6}, \frac{5\pi}{6}$
- $\sin\theta = -1 \ggg \theta = \frac{3\pi}{2}$

9. $3\sec^2\theta - 2\tan^2\theta - 4 = 0$

Answer: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

 **Solution** >

Use $\sec^2\theta = 1 + \tan^2\theta$ and substitute $X = \tan\theta$:

$$3(1 + X^2) - 2X^2 - 4 = 0 \ggg X^2 - 1 = 0$$

Therefore $X = \pm 1$.

- $\tan\theta = 1 \ggg \theta = \frac{\pi}{4}, \frac{5\pi}{4}$
- $\tan\theta = -1 \ggg \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$